

Nikkiso Communication Protocol for DBB-27

This document has described the communication protocol for DBB-27 by the Nikkiso CO.

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1. Revision History

Rev.	Description
2019-09-12	New

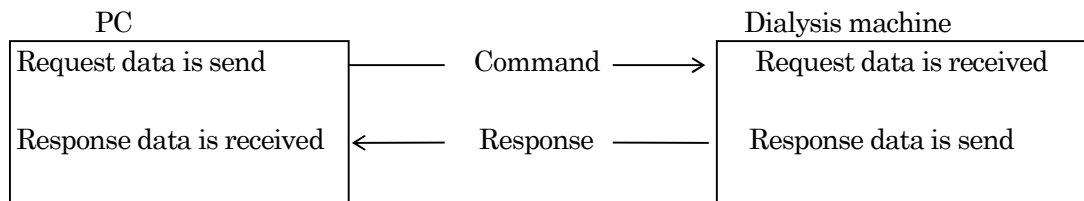
2. Introduction

This document describes the communication protocol between the dialysis machine (DBB-27) with the central management computer (CP).

All communications between Dialysis machine and CP are done following the next process:

CP transmits command to Dialysis machine then, Dialysis machine sends back the response for that command to the CP.

Moreover, in the communication data, ASCII data is used for numerical value.



Recommended Action

If normal reception is not possible even after retrying several times (about 3 times), it is recommended to handle the error on the PC side

3. Interface

Dialysis machine have 2 interface.

- 1) DSUB connector : JIS, Insulated type
- 2) RJ45 (TCP / IP)

Communication Specification (DSUB connector is serrected)

- 1) Communication : RS232C (Async / half duplex)
- 2) Speed : 9,600bps
- 3) Character set : ASCII
- 4) Data Length : 8bit
 - Stop bit : 1bit
 - Parity : none
 - Xon/Xoff : none
- 5) Cable : If you use a commonly used RS232C cable, use a 9-pin cross cable.
- 6) Pin location : See Table-1

<Table-1> Pin location

Pin No	Signal name
1	CD
2	RXD
3	TXD
4	DTR
5	GND
6	DSR
7	RTS
8	CTS
9	RI

4. Format

In the following explanation, characters in “” are converted to ASCII code at data sending.

4.1 from PC to Dialysis machine (command)

Request data (from CP to dialysis machine) is 3byte as follow;

“K”	CR	LF
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4.2 from Dialysis machine to PC (response)

Data (from dialysis machine to CP) is variable length as follow;

STX	LEN	RES-DATA	SUM	ETX
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STX	Data start code [2bytes] "K"
LEN	"2" Version number Data Length [3bytes] Byte size of all "ID" and "DATA" added 3 decimal digits Ex.) In case of a total of 123Byte; "1", "2", "3"
RES-DATA	Response data [variable length] See Table-2 Transmitted Data
SUM	Check sum data[2bytes] All of data size are added except "CR", "LF", then convert last 2digits (HEX) to ASCII code. Ex.)In case of a total of 5aH Byte; "5", "a"
ETX	Data end code[2bytes] CR, LF

5. Appendix

- (1) The data from No.1 to 12 is 5 decimal digits with a decimal point.
The first-digit is “-” when the data is below zero.
Ex.) In case of Dialysate flow pressure [-146mmHg],
“-”, “0”, “1”, “4”, “6”
High byte is “0” when high byte of the data is 0.
Ex.) In case of UF goal [2.35L]
“0”, “2”, “.”, “3”, “5”
- (2) Treatment time is less than 479 min.
- (3) The data from No.13 to 20 and 31: alarm condition is “1” and false is “0”.
- (4) The data No.20 (Other alarm) is all of alarm except the alarm from No.13 to 19 and 31.
- (5) The data No.22 (Under treatment flag), Under treatment (Connection, Treatment and Disconnect mode) is “1” and false is “0”.
- (6) Regarding treatment mode, HD is “0”, ECUM is “1”
- (7) Blood pressure measurement time is 24-hour notation.
- (8) The last measurement result is transmitted at the data from No.27 to 30.

<Table-2> Transmitted data

No.	Data name	Data ID code	Data size	Units	Remark
1	UF goal	"A"	HEX5Byte	L	Set Value
2	UF volume	"B"	HEX5Byte	L	Actual value
3	UF rate	"C"	HEX5Byte	L/hr	Actual value
4	Blood pump flow rate	"D"	HEX5Byte	mL/min	Actual value
5	Infusion pump rate (Heparin pump)	"E"	HEX5Byte	mL/hr	Set Value
6	Dialysate temperature	"F"	HEX5Byte	Deg C	Actual value
7	Dialysate conductivity	"G"	HEX5Byte	mS/cm	Actual value
8	Venous pressure	"H"	HEX5Byte	mmHg	Actual value
9	Dialysate pressure	"I"	HEX5Byte	mmHg	Actual value
10	TMP	"J"	HEX5Byte	mmHg	Actual value
11	Treatment time	"K"	HEX5Byte	min	
12	Dialysate flow rate	"L"	HEX5Byte	mL/min	
13	Dialysate temperature alarm	"a"	HEX1Byte		"0": No Alarm "1": Alarm
14	Conductivity alarm	"b"	HEX1Byte		"0": No Alarm "1": Alarm
15	Venous pressure alarm	"c"	HEX1Byte		"0": No Alarm "1": Alarm
16	Dialysate pressure alarm	"d"	HEX1Byte		"0": No Alarm "1": Alarm
17	TMP alarm	"e"	HEX1Byte		"0": No Alarm "1": Alarm
18	Air alarm	"f"	HEX1Byte		"0": No Alarm "1": Alarm
19	Blood leak alarm	"g"	HEX1Byte		"0": No Alarm "1": Alarm
20	Other alarm	"h"	HEX1Byte		"0": No Alarm "1": Alarm
21	Under treatment flag	"M"	HEX1Byte		
22	Treatment mode	"N"	HEX1Byte		"0": No Treatment "1": Treatment
23	Substitution goal	"O"	HEX5Byte	L	Output 0x30 because of nothing target item.
24	Substitution volume	"P"	HEX5Byte	L	Output 0x30 because of nothing target item.
25	Substitution rate	"Q"	HEX5Byte	L/hr	Output 0x30 because of nothing target item.
26	Substitution temperature	"R"	HEX5Byte	Deg C	Output 0x30 because of nothing target item.
27	Blood pressure measurement time	"S"	HEX5Byte	HHMMSS	
28	Systolic blood pressure	"T"	HEX5Byte	mmHg	
29	Diastolic blood pressure	"U"	HEX5Byte	mmHg	
30	Pulse	"V"	HEX5Byte	bpm	
31	Blood pressure alarm	"i"	HEX1Byte		"0": No Alarm "1": Alarm